

NLP Today – Session 1 Report

Part-of-Speech Tagging and Constituency Parsing

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The dataset has four general-domain sentences built around classic ambiguities and two specialty-domain sentences (biology, astronomy). Section 1 annotates them by hand (PoS tags and the constituency trees we want). Section 2 compares two automatic PoS taggers, parses the sentences with the provided CYK implementation under several context-free grammars and PoS inputs, and compares the result with a neural constituency parser. Section 3 discusses the results. Every number and automatic tree in this report is produced by our scripts (Appendix A); `cyk.py` is used unmodified.

1 Manual analysis

1.1 Annotation conventions

- **Tokens.** Words and punctuation marks are split, but lexical units are kept whole: the abbreviation *Dec.*, the decimal numbers *19.33* and *0.044*, the compound *type-Ia*, and the catalogue name *J10354824+3900279* (a “Possible SuperNova” designation built from its sky coordinates).
- **Tags.** We use the imposed Universal Dependencies (UD) PoS list ADJ ADP AUX DET NOUN NUM PROPN PUNCT SCONJ VERB. Following UD, copular and passive *be* are AUX, *when/that* introducing a clause are SCONJ, and *like/as* before a noun phrase are ADP.
- **Exception.** *they*, *which* and *this* are pronouns, and the list has no PRON tag. Forcing DET or NOUN would put a wrong label into the reference that every evaluation relies on, so we tag them PRON (marked †) and treat the gap as a limitation of the tag set.
- **Convention-dependent choices.** The species name *Arabidopsis thaliana* is tagged PROPN (it names a unique taxon; Penn-Treebank-style corpora use a common-noun tag). *UT* (Universal Time) is tagged PROPN, like *GMT*. The variable *z* is tagged NOUN, because SYM is not available. *type-Ia* is tagged NOUN: it heads the elliptical noun phrase *a type-Ia [supernova]*.

1.2 Gold part-of-speech annotation

- S1 (*general*) The_{DET} cat_{NOUN} sat_{VERB} on_{ADP} the_{DET} couch_{NOUN} .PUNCT
- S2 (*general*) Time_{NOUN} flies_{VERB} like_{ADP} an_{DET} arrow_{NOUN} .PUNCT
- S3 (*general*) The_{DET} spy_{NOUN} saw_{VERB} the_{DET} cop_{NOUN} with_{ADP} the_{DET} telescope_{NOUN} .PUNCT
- S4 (*general*) The_{DET} spy_{NOUN} saw_{VERB} the_{DET} cop_{NOUN} with_{ADP} the_{DET} revolver_{NOUN} .PUNCT
- S5 (*biology*) Arabidopsis_{PROPN} thaliana_{PROPN} seedlings_{NOUN} exhibit_{VERB} longer_{ADJ} hypocotyls_{NOUN} when_{SCONJ} they_{PRON†} are_{AUX} grown_{VERB} under_{ADP} high_{ADJ} ambient_{ADJ} temperature_{NOUN} ,PUNCT which_{PRON†} is_{AUX} defined_{VERB} as_{ADP} thermomorphogenesis_{NOUN} .PUNCT
- S6 (*astronomy*) A_{DET} spectrogram_{NOUN} of_{ADP} PSN_{PROPN} J10354824+3900279_{PROPN} obtained_{VERB} on_{ADP} Dec._{PROPN} 19.33_{NUM} UT_{PROPN} suggests_{VERB} that_{SCONJ} this_{PRON†} is_{AUX} a_{DET} type-Ia_{NOUN} at_{ADP} redshift_{NOUN} z_{NOUN} 0.044_{NUM} .PUNCT

1.3 Ambiguity analysis

Out of context, many word forms can take several PoS tags (Table 1). These alternatives are fed to the parser in §2.2.

Structural ambiguities, and the reading we choose as target:

- **S2** *Time flies like an arrow* has four readings: (a) time passes as quickly as an arrow; (b) imperative: time (measure) the flies that resemble an arrow; (c) *time-flies* (a kind of fly) are fond of an arrow; (d) imperative: time the flies the way you would time an arrow. Only (a) is natural and it is our target. Reading (c) changes the tags (*flies*/NOUN, *like*/VERB); (b) and (d) need *Time*/VERB.

Table 1: Words that can take another PoS out of context (gold tag / alternatives).

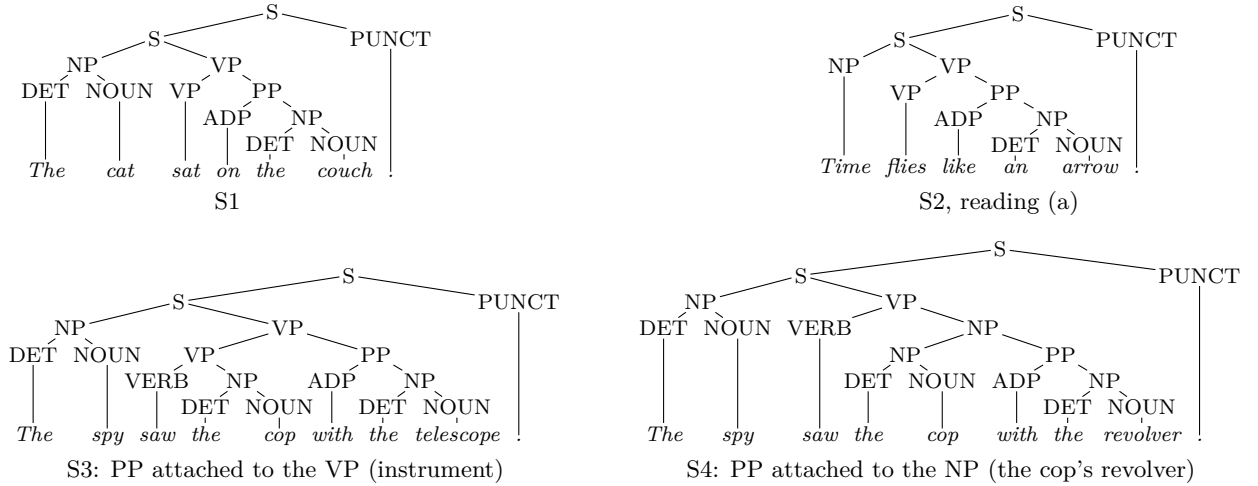
S1	<i>couch</i> : NOUN / VERB
S2	<i>Time</i> : NOUN / VERB; <i>flies</i> : VERB / NOUN; <i>like</i> : ADP / VERB
S3	<i>spy</i> : NOUN / VERB; <i>saw</i> : VERB / NOUN; <i>cop</i> : NOUN / VERB; <i>telescope</i> : NOUN / VERB
S4	<i>spy</i> : NOUN / VERB; <i>saw</i> : VERB / NOUN; <i>cop</i> : NOUN / VERB
S5	<i>Arabidopsis</i> : PROPN / NOUN; <i>thaliana</i> : PROPN / NOUN; <i>exhibit</i> : VERB / NOUN; <i>grown</i> : VERB / ADJ; <i>high</i> : ADJ / NOUN; <i>defined</i> : VERB / ADJ; <i>as</i> : ADP / CONJ
S6	<i>obtained</i> : VERB / ADJ; <i>UT</i> : PROPN / NOUN; <i>that</i> : CONJ / DET, PRON; <i>this</i> : PRON / DET; <i>type-Ia</i> : NOUN / ADJ

- **S3/S4.** The phrase *with the telescope/revolver* can modify the verb (the instrument of seeing) or the object (the cop who holds it). World knowledge selects the verb for the telescope and the object for the revolver. Yet both sentences have the *same* PoS sequence.
- **S5.** The *when*-clause modifies *exhibit longer hypocotyls*. The relative clause *which is defined as thermomorphogenesis* could attach to the closest noun phrase *high ambient temperature*. However, thermomorphogenesis is the growth response to warmth, not the temperature itself, so the clause refers to the whole preceding statement.
- **S6.** *obtained on Dec. 19.33 UT* modifies *a spectrogram of PSN J...* (a spectrogram is obtained, not a supernova). *at redshift z 0.044* can attach to *a type-Ia* or to the predicate *is a type-Ia*; both mean the same.

1.4 Target constituency trees

The labels are limited to S, NP, VP, PP plus the PoS tags. The trees are drawn directly in the binary form that CYK can produce (§2.2):

- a one-word NP/VP is a single lexical node, e.g. (NP Time);
- pre-modifiers attach right-branching (NP $\rightarrow$ ADJ NP);
- embedded clauses reuse S, since SBAR is not allowed.

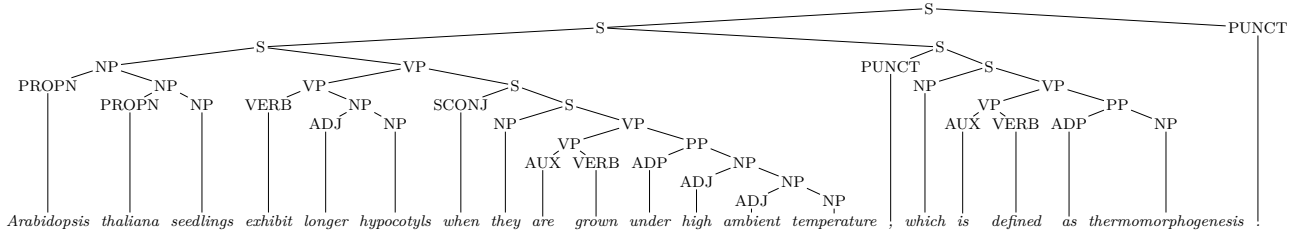
**Figure 1:** Target trees for the general-domain sentences.

2 Automatic analysis

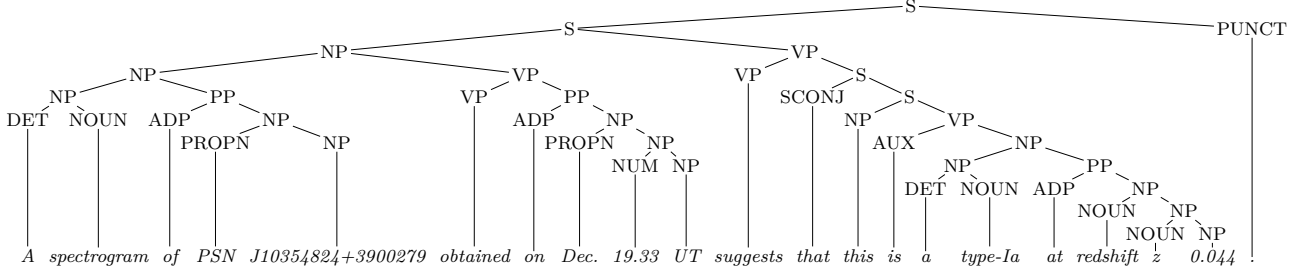
2.1 PoS tagging: spaCy vs. sciSpaCy

Models. We compare two models, both running on spaCy 3.7.5:

- `en_core_web_sm` 3.7.1, trained on OntoNotes 5 (reported tag accuracy 97.2%);
- `sciSpaCy en_core_sci_sm` 0.5.4, trained on GENIA 1.0 (MEDLINE abstracts about human blood cells and transcription factors) and OntoNotes 5.



S5 (relative clause attached to the whole clause)



S6 (obtained... modifies A spectrogram of PSN...; at redshift... modifies a type-Ia)

Figure 2: Target trees for the specialty-domain sentences.

Both models predict fine-grained Penn Treebank (PTB) tags. The universal tag is then derived by rules (spaCy’s `attribute_ruler`). Tokens are aligned with the gold tokens by character offsets; a token that a model splits or merges counts as an error.

Table 2: Tagging accuracy against the manual annotation (73 tokens).

Model	General (S1–S4)	Specialty (S5–S6)	All	Tokenisation mismatches
en_core_web_sm 3.7.1	31/31 (100.0%)	36/42 (85.7%)	67/73 (91.8%)	2
en_core_sci_sm 0.5.4	30/31 (96.8%)	34/42 (81.0%)	64/73 (87.7%)	1

Table 3: Every token where a model differs from the gold tag (errors in red; the PTB tag is in subscript).

Token	Gold	en_core_web_sm	en_core_sci_sm
S2 <i>flies</i>	VERB	VERB _{VBZ}	NOUN _{NNS}
S5 <i>Arabidopsis</i>	PROPN	NOUN _{NN}	NOUN _{NN}
S5 <i>thaliana</i>	PROPN	NOUN _{NN}	NOUN _{NN}
S5 <i>longer</i>	ADJ	ADV _{RBR}	ADV _{RBR}
S5 <i>hypocotyls</i>	NOUN	ADJ _{JJ}	ADJ _{JJ}
S5 <i>which</i>	PRON	PRON _{WDT}	DET _{WDT}
S6 <i>PSN</i>	PROPN	PROPN _{NNP}	NOUN _{NN}
S6 <i>J10354824+3900279</i>	PROPN	J10354824 _{PROPN/NNP} +NOUN/NN 3900279 _{NUM/CD}	J10354824 _{NOUN/NN} +CCONJ/CC 3900279 _{NUM/CD}
S6 <i>this</i>	PRON	PRON _{DT}	DET _{DT}
S6 <i>type-Ia</i>	NOUN	type _{NOUN/NN} -PUNCT/HYPH Ia _{NOUN/NN}	NOUN _{NN}

What the differences show:

- General domain.** Both models are almost perfect. The only error is sciSpaCy tagging *flies* as NOUN (the *time-flies* reading). That choice is plausible locally, but combined with *like*/ADP the sentence is left without any verb (see §2.2).
- Unknown domain word.** Both models read the plant term *hypocotyls* as an adjective and then *longer* as an adverb (*exhibit longer*). One unknown word produces two errors. Being “biomedical” does not help sciSpaCy here: plant morphology and astronomy are outside GENIA.
- Tokenisation.** Both tokenisers split *J10354824+3900279*, and sciSpaCy even tags *+* as a coordinating conjunction. spaCy also splits *type-Ia*; sciSpaCy’s tokeniser, adapted to hyphenated biomedical terms, keeps it whole.
- Conventions, not errors.** Both models tag *Arabidopsis thaliana* as a common noun (the PTB/GENIA convention). This disagrees with our PROP choice without either side being wrong.

5. **which/this: a pipeline effect.** Both models predict the *same* PTB tags (WDT, DT), yet their universal tags differ. The mapping rule “DT/WDT whose dependency relation is not `det` $\rightarrow$ `PRON`” needs the parser’s output. In `en_core_web_sm` the `attribute_ruler` runs *after* the parser, so the rule fires (`PRON`). In `en_core_sci_sm` it runs *before* the parser, so it never fires (`DET`). The difference comes from component order, not from the statistical model.

With 73 tokens, one token is worth 1.4 points of accuracy, so these results should be read qualitatively.

2.2 Constituency parsing with CYK

The implementation.

- `cyk.py` receives a *list* of candidate tags for each word, so lexical ambiguity can be passed to the parser.
- It accepts only binary rules $X \rightarrow YZ$ and lexical rules $X \rightarrow tag$ (Chomsky normal form, CNF). It has no unary rule $X \rightarrow Y$, so unit rules are eliminated as in CNF conversion: $NP \rightarrow NOUN \mid PROPEN \mid PRON \mid NUM$ and $VP \rightarrow VERB$ become lexical rules.
- It builds every tree explicitly and returns *all* constituents covering the input, whatever their label; we keep those labelled `S`.
- To avoid materialising millions of trees, we first count them with the same dynamic programme.
- Sanity check: with the grammar of its docstring, it reproduces the documented tree for *The cat sat on the couch*.

Grammars. Every grammar contains a lexicon mapping each tag to itself, plus the unit rules $NP \rightarrow NOUN$, $NP \rightarrow PROPEN$, $NP \rightarrow PRON$, $NP \rightarrow NUM$, $VP \rightarrow VERB$.

G_general (8 binary rules)

$S \rightarrow NP VP$, $S \rightarrow S PUNCT$, $NP \rightarrow DET NOUN$, $NP \rightarrow NOUN NP$, $NP \rightarrow NP PP$, $PP \rightarrow ADP NP$, $VP \rightarrow VERB NP$, $VP \rightarrow VP PP$.

G_general+imp

`G_general` + the CNF image of the imperative rule $S \rightarrow VP$: $S \rightarrow VERB NP$, $S \rightarrow VP PP$.

G_full (19 binary rules)

`G_general` + 11 rules for the specialty sentences: $NP \rightarrow ADJ NP$, $NP \rightarrow PROPEN NP$, $NP \rightarrow NUM NP$, $NP \rightarrow NP VP$, $NP \rightarrow NP S$, $VP \rightarrow AUX VERB$, $VP \rightarrow AUX NP$, $VP \rightarrow VP S$, $S \rightarrow SCONJ S$, $S \rightarrow PUNCT S$, $S \rightarrow S S$.

G_treebank

The rules read off our six target trees, i.e. `G_full` without $NP \rightarrow NP S$, which no target uses.

We avoid *spurious* ambiguity (the same analysis built twice). Since $VP \rightarrow VERB$ is lexical, no grammar has both $X \rightarrow VERB Y$ and $X \rightarrow VP Y$. We use $VP \rightarrow AUX VERB$ rather than $VP \rightarrow AUX VP$, which would build both `[are [grown under...]]` and `[[are grown] under...]`. For the same reason we use $NP \rightarrow DET NOUN$ rather than $NP \rightarrow DET NP$.

PoS inputs.

- *manual*: the gold tags;
- *spaCy* and *sciSpaCy*: the tags each model outputs, on its own tokens;
- *ambiguous*: for each word, the union of the gold tag, the alternatives of Table 1 and both taggers’ tags.

Time flies like an arrow. The ambiguous input is *Time*: {`NOUN`, `VERB`}, *flies*: {`VERB`, `NOUN`}, *like*: {`ADP`, `VERB`}, *an*: {`DET`}, *arrow*: {`NOUN`}.

- `G_general` returns readings (a) and (c).
- Adding the two imperative rules returns *exactly* the four readings of §1.3 (Fig. 3), including the three trees of the lecture’s Chomsky example.
- `G_full` returns three trees instead. The extra one, (`S` (`NP` (`NP` *Time*) (`VP` *flies*)) (`VP` *like an arrow*)) (“time [that] flies like(s) an arrow”), comes from the reduced-relative rule $NP \rightarrow NP VP$ added for `S6`. It is not English: the verb agreement is wrong, and a reduced relative needs a participle. But a context-free grammar over universal tags sees neither agreement nor finiteness, since `VBZ` and `VBN` both become `VERB`.
- Adding the imperative rules to `G_full` gives 23 trees: rules added for one sentence interact with all the others.

Table 4: Number of S-rooted trees returned by CYK (final period kept). ✓ and bold: the target tree is among them. *: counted but too many trees to build (the target is derivable, since it is found with the smaller G_treebank).

	PoS input	G_general	G_general+imp	G_full	G_treebank
S1	manual	1 ✓	1 ✓	1 ✓	1 ✓
	spaCy	1 ✓	1 ✓	1 ✓	1 ✓
	sciSpaCy	1 ✓	1 ✓	1 ✓	1 ✓
	ambiguous	1 ✓	1 ✓	1 ✓	1 ✓
S2	manual	1 ✓	1 ✓	1 ✓	1 ✓
	spaCy	1 ✓	1 ✓	1 ✓	1 ✓
	sciSpaCy	0	0	0	0
	ambiguous	2 ✓	4 ✓	3 ✓	3 ✓
S3	manual	2 ✓	2 ✓	2 ✓	2 ✓
	spaCy	2 ✓	2 ✓	2 ✓	2 ✓
	sciSpaCy	2 ✓	2 ✓	2 ✓	2 ✓
	ambiguous	2 ✓	2 ✓	2 ✓	2 ✓
S4	manual	2 ✓	2 ✓	2 ✓	2 ✓
	spaCy	2 ✓	2 ✓	2 ✓	2 ✓
	sciSpaCy	2 ✓	2 ✓	2 ✓	2 ✓
	ambiguous	2 ✓	2 ✓	2 ✓	2 ✓
S5	manual	0	0	633 ✓	97 ✓
	spaCy	0	0	0	0
	sciSpaCy	0	0	0	0
	ambiguous	0	0	5064 ✓	776 ✓
S6	manual	0	0	557 ✓	300 ✓
	spaCy	0	0	0	0
	sciSpaCy	0	0	0	0
	ambiguous	0	0	4632 *	2072 ✓

- With a single tag per word, the manual and spaCy tags yield only the intended tree (the tagger has already chosen the reading), whereas sciSpaCy’s tags yield *no* tree.

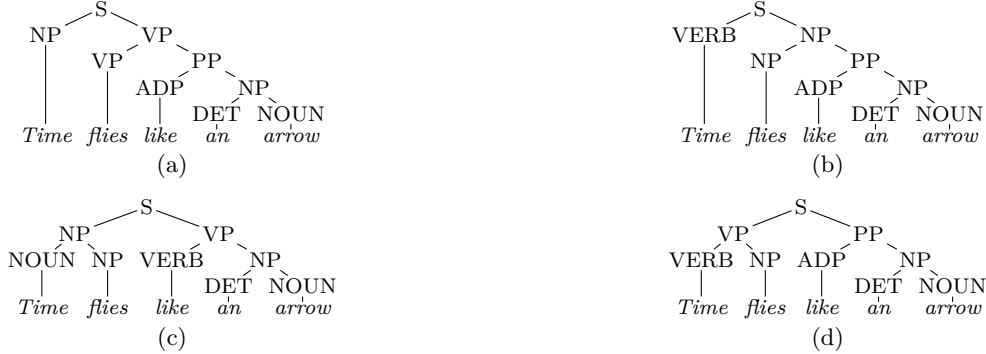


Figure 3: The four trees of *Time flies like an arrow* returned by G_general+imp (final period omitted). Readings as in §1.3.

The spy sentences. Every grammar and every input gives exactly two trees for S3 and for S4: PP attached to the VP (Fig. 1, S3) and PP attached to the NP (S4). The two sentences share the PoS sequence DET NOUN VERB DET NOUN ADP DET NOUN, so *any* grammar over PoS tags returns the same set of trees for both. No single context-free grammar over tags can give the instrument reading for *telescope* and the possession reading for *revolver*: that requires lexical information, e.g. a lexicalised probabilistic grammar estimating how likely a *with*-phrase headed by *telescope* or *revolver* is to attach to *saw* or to *cop*. Deleting one attachment rule to force a reading would break the other sentence.

The specialty sentences. G_general cannot parse S5 or S6: it lacks rules for adjectives and proper nouns as modifiers, auxiliaries and subordinate clauses. G_full derives both targets, but among 633 and 557 trees. The ambiguity breaks down as follows.

- **Punctuation (spurious).** $S \rightarrow S \text{ PUNCT}$ can attach the final period to any clause that ends the sentence, e.g. inside the relative clause. Removing the period halves the counts (Table 5) without changing any phrase.
- **Coverage vs. ambiguity.** $NP \rightarrow NP S$ (a relative clause modifying a noun) is needed in general English, but no target uses it here. Removing it (G_treebank) leaves 97 and 300 trees.
- **Genuine attachment decisions.** Without the final period, G_treebank yields 38 trees for S5. They agree on every local phrase and differ only on where the *when*-clause attaches (VP or clause), where the relative clause attaches (main clause, main VP, *when*-clause, or *they are grown...*), on which side the comma goes, and whether *under...* and *as...* attach low or high. The 116 trees for S6 differ on which NP *obtained...* modifies, on how the date *Dec. 19.33 UT* is bracketed (some trees even read *19.33 UT suggests...* as a clause), on whether *that...* attaches to the VP or to the clause, and on where *at redshift...* attaches. The number of trees is roughly the product of the choices at each decision point, so it grows exponentially with sentence length. With the ambiguous input, the counts reach 5064 and 4632.
- **Tagger input: no tree at all.** In S5, *longer*/ADV is covered by no rule, and *hypocotyls*/ADJ cannot head a noun phrase. In S6, the split-off tokens + (from *J10354824+3900279*) and, for spaCy, the hyphen of *type-Ia* appear in no rule, and sciSpaCy’s *this*/DET and *which*/DET cannot form a noun phrase. One wrong tag or token is enough to lose the whole analysis, and CYK returns no partial result.

Table 5: Effect of the sentence-final period on the number of trees (manual tags).

	Grammar	Period kept	Period removed
S5	G_full	633	315
S5	G_treebank	97	38
S6	G_full	557	230
S6	G_treebank	300	116

2.3 Comparison with a neural constituency parser

We use the Berkeley Neural Parser (benepar 0.2.0, model benepar_en3: T5-small encoder, 95.40 F1 on the revised WSJ test set) through its spaCy plugin. We parse both the raw text and our gold tokens: S1–S5 get identical trees, and S6 differs only by spaCy’s token splits. Table 6 compares the decisive points with our targets.

Table 6: Key decisions: our target vs. benepar_en3.

	Decision	Target (§1.4)	benepar	Same?
S1	whole tree	–	same bracketing	✓
S2	reading	(a) time passes	(a)	✓
S3	<i>with the telescope</i>	attached to VP (instrument)	attached to VP	✓
S4	<i>with the revolver</i>	attached to NP (possession)	attached to VP	×
S5	<i>when</i> -clause	attached to VP <i>exhibit...</i>	attached to VP	✓
S5	relative clause <i>which...</i>	whole clause	NP <i>high ambient temperature</i>	×
S5	<i>Arabidopsis thaliana seedlings</i>	right-branching (grammar limit)	[[<i>Arabidopsis thaliana</i>] seedlings] (NML)	better
S6	<i>obtained on...</i>	<i>A spectrogram of PSN...</i>	same	✓
S6	<i>at redshift $z\ 0.044$</i>	attached to NP <i>a type-Ia</i>	attached to VP <i>is...</i>	≈
S6	spaCy tokens (+, <i>type - Ia</i>)	CYK: no tree	correct tree	–

benepar always outputs a single tree and uses richer labels (SBAR, WHNP, NML). Tagging jointly with the whole sentence, it gets *longer*/JJR and *hypocotyls*/NNS right, where both spaCy taggers failed. Its errors come from treebank preferences rather than meaning: the same attachment for *telescope* and *revolver*, and attachment of the relative clause to the closest noun phrase.

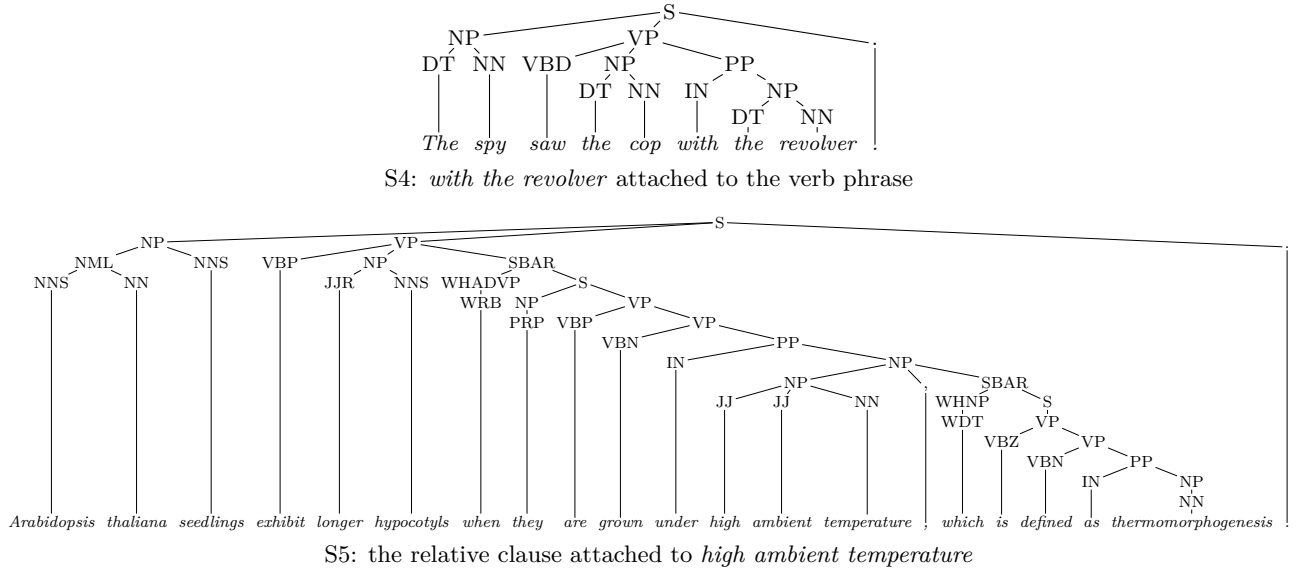


Figure 4: Two `benepar_en3` trees (Penn Treebank labels) that differ from our targets.

3 Observations

3.1 PoS tagging approaches

	Strengths	Weaknesses
Manual	Uses meaning and world knowledge (elliptical <i>type-Ia</i> ; <i>which</i> referring to a clause); convention choices are explicit and documented.	Slow, does not scale; needs written conventions. The imposed tag set (no PRON, ADV, CCONJ, SYM) forces compromises, and some decisions (PROPN vs. NOUN for species) are conventions on which corpora disagree, so “accuracy” partly measures agreement on conventions.
spaCy (web_sm)	Fast; 100% on general sentences; resolves lexical ambiguity from context (<i>flies</i> /VERB).	Unknown domain words cause chains of errors (<i>hypocotyls</i> → ADJ, then <i>longer</i> → ADV); the generic tokeniser splits identifiers and hyphenated terms; universal tags are converted from PTB tags by rules.
sciSpaCy (sci_sm)	Domain tokeniser (<i>type-Ia</i> kept whole); scientific vocabulary.	“Biomedical” ≠ all science: no gain on plant biology or astronomy (81.0% vs. 85.7%), and worse on general text; its tag conversion runs before the parser, so pronoun DT/WDT become DET.

3.2 Grammar-based parsing with CYK

1. **The grammar is the model.** Each rule is a linguistic hypothesis. With 19 binary rules and 5 unit lexical rules, all six targets are derivable.
2. **Coverage costs ambiguity.** Every rule added for one sentence multiplies the trees of the others: with the imperative rules, S2 grows from 3 to 23 trees under `G_full`, and S5 grows from 97 to 633 with `NP` → `NP S`. CYK cannot rank the trees, only enumerate them, and this implementation stores all of them, so memory grows exponentially. A practical parser keeps back-pointers and selects the best tree with a probabilistic grammar (PCFG) and the Viterbi algorithm.
3. **CNF and the restricted labels shape the trees.** Without unary rules, one-word NPs/VPs become lexical nodes. Without an intermediate noun level, NML or SBAR, compounds can only branch to the right (*Arabidopsis thaliana seedlings* should branch left), clauses must reuse S, and some ambiguity is spurious (position of punctuation).

4. **Universal tags are too coarse for syntax.** Finite and participle verbs share VERB, and there are no agreement features, so the grammar over-generates (*Time [that] flies like an arrow*). Agreement cannot be expressed context-free without multiplying the symbols.
5. **No lexical semantics.** A grammar over PoS tags gives identical outputs for S3 and S4.
6. **Fragile pipeline.** A single tagging or tokenisation error leaves no tree (S2 with sciSpaCy; S5 and S6 with both taggers). Passing several candidate tags restores the target but multiplies the trees (up to 5064).

3.3 Grammar-based vs. neural parsing

The neural parser is robust: it always returns a tree, copes with unknown words and different tokenisations, and matches our targets on most decisions. But it returns a single tree without signalling that the sentence is ambiguous, and its attachment choices reflect treebank statistics, not meaning (S4, and the relative clause in S5). CYK with a hand-written grammar has the opposite profile: every ambiguity is explicit, but it ranks nothing and breaks on a single wrong input. Scoring the CYK chart with probabilities (a PCFG or a neural scorer) and returning the n best trees would combine the robustness of the neural parser with the explicit ambiguities of the chart.

A Reproducibility

Python 3.11; spaCy 3.7.5 with `en_core_web_sm` 3.7.1; scispacy 0.6.2 with `en_core_sci_sm` 0.5.4; benepar 0.2.0 (`benepar_en3`) with torch 2.14 (CPU) and transformers 4.46.3; nltk. Two environment issues and their fixes:

- benepar 0.2.0 calls a tokenizer method removed in transformers 5, so transformers is pinned to 4.x.
- sentencepiece cannot open a model file under a Windows path containing non-ASCII characters, so the model is copied to an ASCII folder.

Scripts: `data.py` (dataset, gold tags, target trees), `pos_tagging.py`, `grammar.py` (grammars and tree counting), `cyk_experiments.py`, `benepar_parse.py`, `report_assets.py` (all tables and trees of this report).